WEATHER CONDITIONS AND CLIMATE CHANGE WITH CLIMATEWINS

Priya Agrawal
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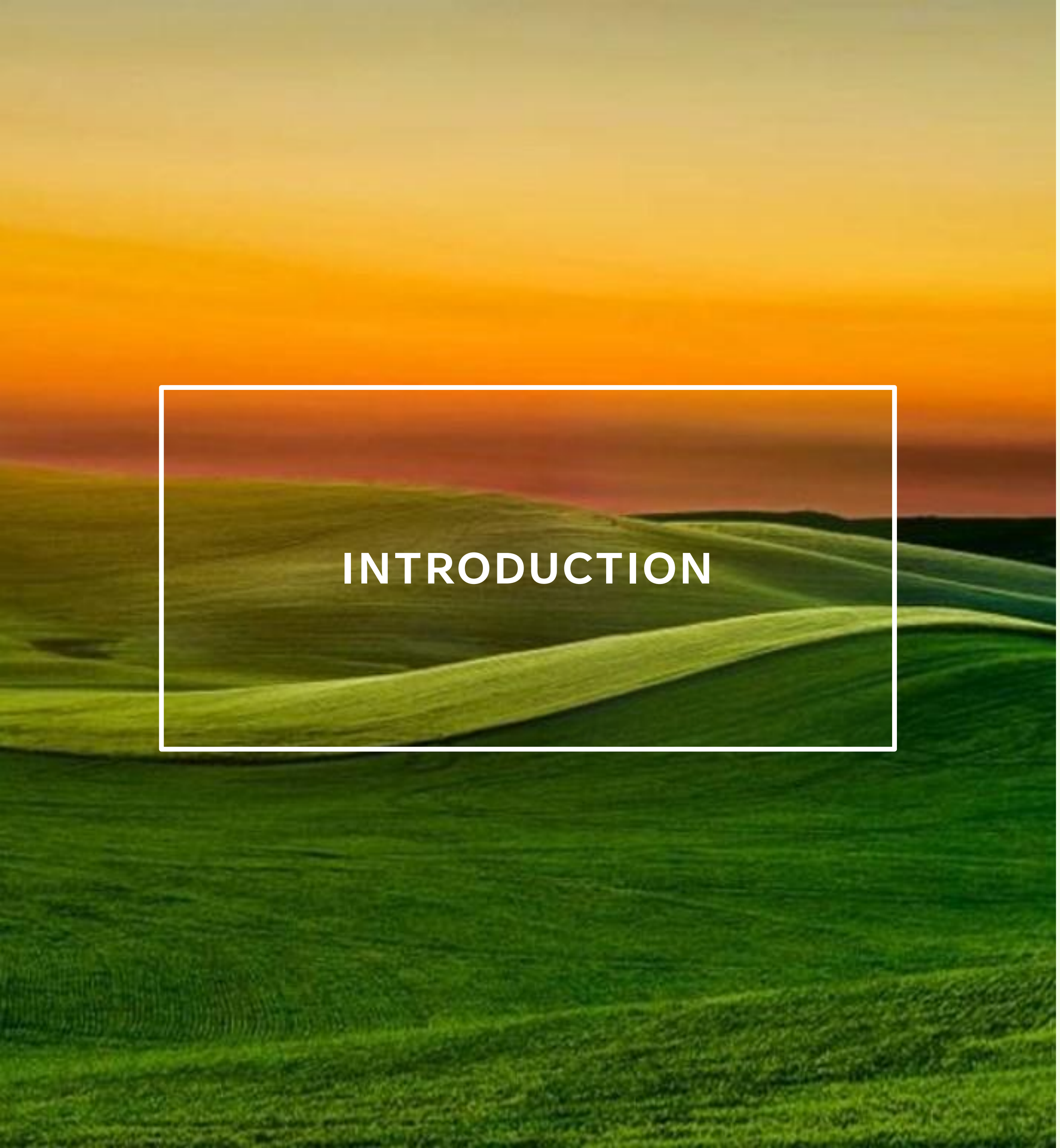

INTRODUCTION

- Over the past 10-20 years, there has been an increase in extreme weather events due to climate change, which is especially of concern to ClimateWins
- ClimateWins aims to use machine learning to explore the following:
  - Identify weather patterns that deviate from Europe’s normal weather patterns
  - Ascertain if there is an increase in frequency of unusual weather patterns
  - Determine what weather conditions may look like over the next 25-50 years using current trends
  - Identify the safest-to-live areas in Europe over the next 25-50 years

THOUGHT EXPERIMENTS

IDENTIFYING THE SAFEST PLACES IN EUROPE USING RANDOM FORESTS: INTEGRATE DATA ON SEA LEVELS, EXTREME WEATHER EVENTS, AND INJURIES AND DEATHS DUE TO EXTREME WEATHER. MODEL WILL CATEGORIZE REGIONS BY SAFETY FOR THE NEXT 25-50 YEARS.

PREDICTING FUTURE WEATHER CONDITIONS USING LSTM: OBTAIN WEATHER PREDICTIONS FOR THE NEXT 25-50 YEARS BY TRAINING AN LSTM MODEL USING PAST AND CURRENT WEATHER DATA INCLUDING EXTREME WEATHER EVENT DATA.

IDENTIFYING UNUSUAL WEATHER PATTERNS USING HIERARCHICAL CLUSTERING: CURRENT AND PAST WEATHER DATA AND EXTREME WEATHER EVENT DATA CAN BE USED TO IDENTIFY ANY ANOMALIES IN WEATHER PATTERNS ACROSS EUROPE.

MACHINE LEARNING MODELS

1. **Random Forests:**

Made up of many decision trees and each tree is trained on a random sample from the data set. The predictions generated by all the trees are averaged to obtain a final prediction. The model's ability to use both categorical and continuous data will allow it to produce safety classifications for regions.

2. **Long Short-Term Memory (LSTM):**

A type of Recurrent Neural Network which in addition to being able to process sequential data such as time series data by keeping a memory of past data to use it as a base for future predictions, it forgets any data that is not needed. This will allow weather predictions to be made based on weather pattern changes from past to present.

3. **Hierarchical Clustering:**

Individual data points are categorized from the bottom up in the form of an upside-down tree (dendrogram). Each data point is merged into branches depending on similarity. This will allow unusual weather patterns to be seen clearly.

NECESSARY ADDITIONAL DATA

- Sea level data
- Extreme weather event data:
  - Storms
  - Natural disasters
  - Floods
  - Droughts
- Data on injuries and deaths caused by extreme weather events

THOUGHT EXPERIMENT 1

- **Hypothesis:** If regions in Europe have small annual temperature changes, relatively stable sea levels, and small changes in the incidence of extreme weather events, then they are more likely to be safer for people to live in within the next 25-50 years.
- **Objective:** Categorize regions in Europe as being of high, medium, or low safety in terms of climate conditions to identify which regions would be the safest for people to reside in within the next 25-50 years.
- **Machine Learning Method:** Random Forests
- **Data:**
    - Weather data set from 1960-2022
    - Sea level data
    - Extreme weather event data: storms, natural disasters, floods, drought
    - Data on injuries and deaths caused by extreme weather

THOUGHT EXPERIMENT 1

Steps for implementing this thought experiment:

- Collect data on sea levels, extreme weather events, and injuries and deaths caused by extreme weather
- Clean new data, address missing data, and integrate new data with existing weather data
- Create target labels for the data which would include the following: High safety, Medium safety, and Low safety
- Reshape data and split into training and testing sets
- Run Grid Search and Random Search to determine which has the highest score and use the optimized hyperparameters from that approach to train the Random Forests model
- Obtain feature importances to see which weather variables have the greatest impact on safety and analyze results for safety classification of regions

THOUGHT EXPERIMENT 2

- **Hypothesis:** If current weather trends persist, then weather conditions over the next 25-50 years will be more extreme and unstable.
- **Objective:** To use current weather trends to predict weather conditions in Europe for the next 25-50 years.
- **Machine Learning Method:** Long Short-Term Memory (LSTM)
- **Data:**
  - Weather data set from 1960-2022
  - Extreme weather event data: storms, natural disasters, floods, drought

THOUGHT EXPERIMENT 2

Steps for implementing this thought experiment:

- Collect extreme weather event data
- Clean new data, address missing data, and integrate new data with existing weather data
- Reshape data and split into training and testing sets
- Create Keras model, compile model, and run model until it converges
- Run Bayesian Optimization to obtain optimized hyperparameters and run model with these hyperparameters
- Compare model accuracy before and after optimization and use the hyperparameters that produce a higher accuracy
- Generate weather predictions for the next 25-50 years

THOUGHT EXPERIMENT 3

- **Hypothesis:** Abnormal weather patterns are occurring in Europe, across all regions.
- **Objective:** To identify weather patterns that deviate from Europe’s normal weather patterns and point out which regions have more changes in weather patterns.
- **Machine Learning Method:** Hierarchical Clustering
- **Data:**
    - Weather data set from 1960-2022
    - Extreme weather event data: storms, natural disasters, floods, drought

THOUGHT EXPERIMENT 3

Steps for implementing this thought experiment:

- Collect extreme weather events data
- Clean new data, address missing data, and integrate new data with existing weather data
- Scale data
- Create subsets of the data set for different year ranges, seasons, individual weather stations, or groups of weather stations with similar climates
- Generate dendrograms for each subset using the Ward method
- Analyze dendrograms to look for sparse clusters or outliers which represent unusual weather patterns

RECOMMENDATIONS

- Thought experiment 1 which involves using Random Forests to categorize regions in Europe as high, medium, or low safety in terms of climate conditions has the most potential
  - Random Forests prevents overfitting
  - This will not only fulfill ClimateWins' goal of identifying the safest regions for people to live in within the next 25-50 years, but will also give an idea of which regions are affected the most by climate change
  - This model will likely produce the most accurate data as it will likely have the highest accuracy rate
- **Algorithm needed:** Random Forests
- **Data:** Weather data from 1960-2022, sea level data, extreme weather events data including storms, natural disasters, floods, and drought, and data on injuries and deaths caused by extreme weather events
- **Next Steps:** Collect extreme weather event data in Europe, create target labels for data, run Grid Search and Random Search, and implement the Random Forests model using optimized hyperparameters

THANK YOU

Please contact me at priya.agrawal0929@gmail.com if you have any questions.

Take at look at my GitHub for the Python code used in this project.